

Lab 11 Class Grade Calculator

Create a program that reads student grades in labs, exams, and quizzes, and then outputs the overall class grade. Your output should exactly match the output below. User input is in **red**.

Console

```
Enter your lab grades. Enter a negative number to stop:
Lab Grade 1: 85
Lab Grade 2: 90
Lab Grade 3: 78
Lab Grade 4: 88
Lab Grade 5: 92
Lab Grade 6: -1

Enter your exam grades. Enter a negative number to stop:
Exam Grade 1: 75
Exam Grade 2: 82
Exam Grade 3: 79
Exam Grade 4: -1

Enter your quiz grades. Enter a negative number to stop:
Quiz Grade 1: 88
Quiz Grade 2: 92
Quiz Grade 3: -1

Calculating your class grade...

Your overall class grade is 84.11
```

Specifications

- Ask users to input lab grades and store them. Stop reading input when a negative value is provided (do not store the negative value).
- Ask users to input exam grades and store them. Stop reading input when a negative value is provided (do not store the negative value).
- Ask users to input quiz grades and store them. Stop reading input when a negative value is provided (do not store the negative value).
- Average each set of grades and output their class grade using the following formula:
$$\text{class_grade} = (\text{avg_lab_grade} * 0.4) + (\text{avg_exam_grade} * 0.4) + (\text{avg_quiz_grade} * 0.2)$$
- Output the overall class grade with a precision of 2 decimal places.
- An input of 0 is a valid grade.